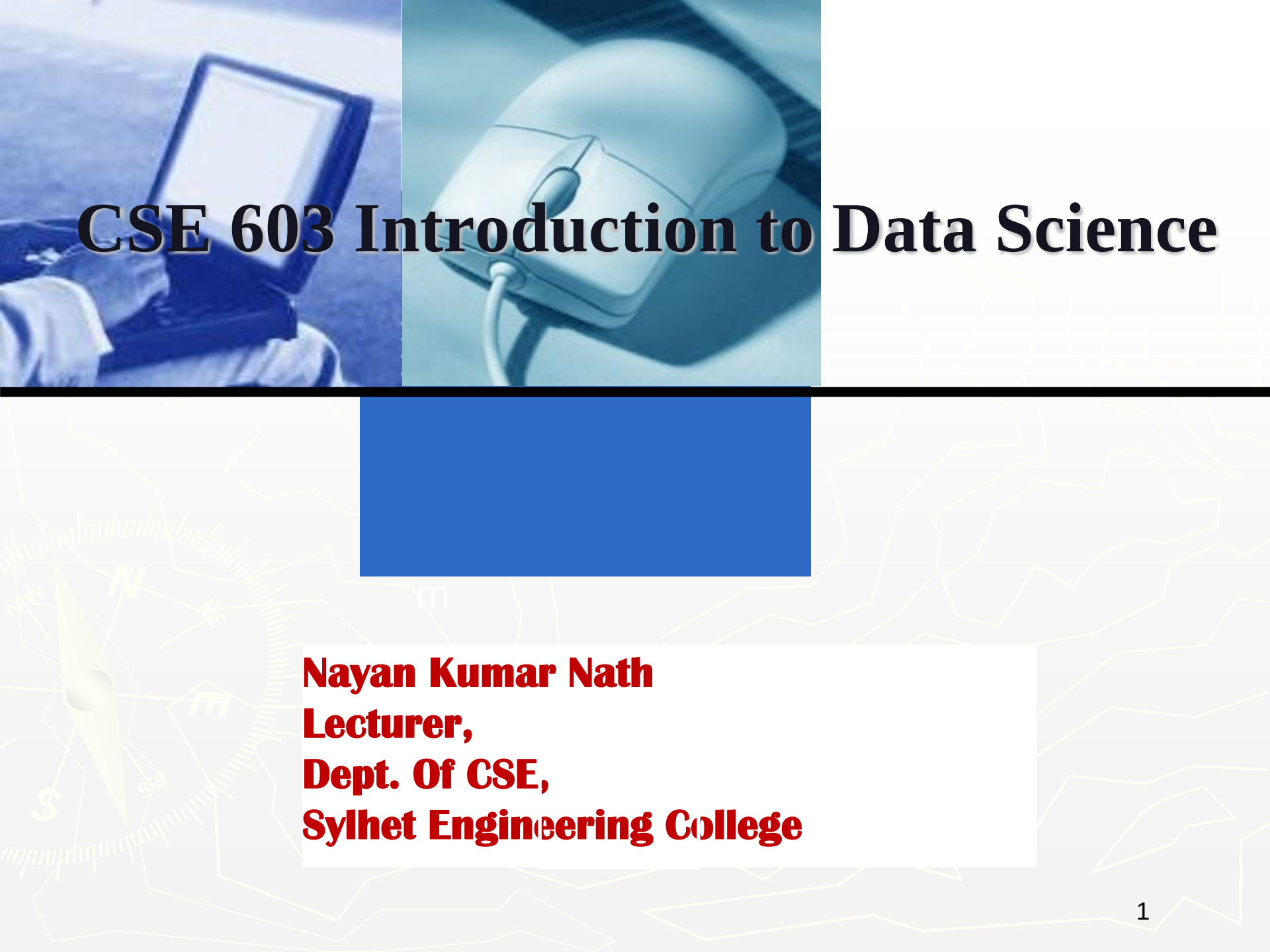




CSE 603 Introduction to Data Science

Nayan Kumar Nath
Lecturer,
Dept. Of CSE,
Sylhet Engineering College

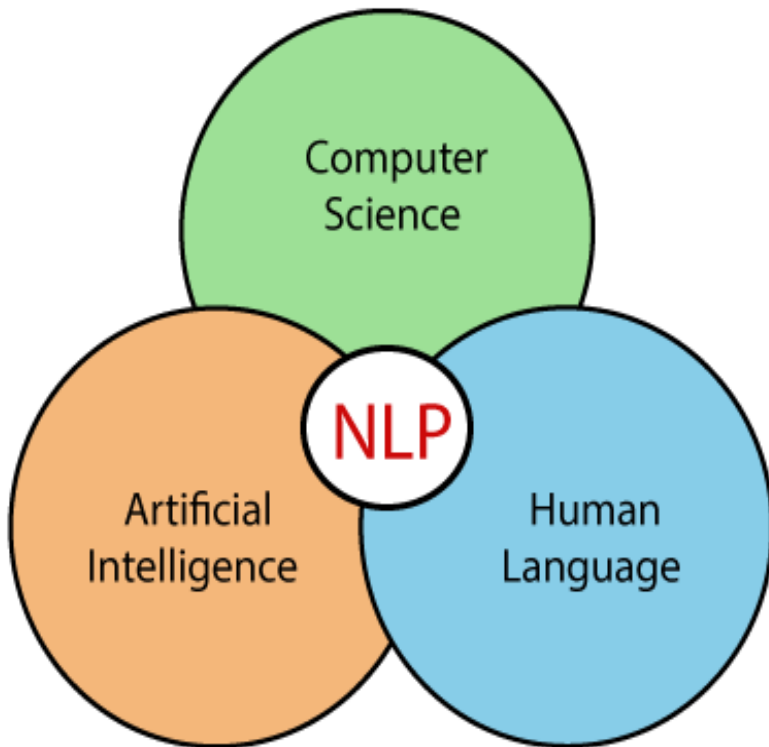


NLP

Natural language
processing

What is Natural Language Processing?

NLP stands for **Natural Language Processing**, which is a part of **Computer Science**, **Human language**, and **Artificial Intelligence**. It is the technology that is used by machines to understand, analyse, manipulate, and interpret human's languages. It helps developers to organize knowledge for performing tasks such as **translation**, **automatic summarization**, **Named Entity Recognition (NER)**, **speech recognition**, **relationship extraction**, and **topic segmentation**. NLP is separate from — but often used in conjunction with — speech recognition, which seeks to parse spoken language into words, turning sound into text and vice versa.



NLP helps machines convert human language into meaningful data and can also generate human-like language.



History of NLP

- **(1940-1960) - Focused on Machine Translation (MT)**

The Natural Languages Processing started in the year 1940s.

- **1948** - In the Year 1948, the first recognizable NLP application was introduced in Birkbeck College, London.
- **1950s** - In the Year 1950s, there was a conflicting view between linguistics and computer science. Now, Chomsky developed his first book syntactic structures and claimed that language is generative in nature.

In 1957, Chomsky also introduced the idea of Generative Grammar, which is rule based descriptions of syntactic structures.

- **(1960-1980) - Flavored with Artificial Intelligence (AI)**

In the year 1960 to 1980, the key developments were:

Augmented Transition Networks (ATN)

Augmented Transition Networks is a finite state machine that is capable of recognizing regular languages.

Case Grammar

Case Grammar was developed by **Linguist Charles J. Fillmore** in the year 1968. Case Grammar uses languages such as English to express the relationship between nouns and verbs by using the preposition.

In Case Grammar, case roles can be defined to link certain kinds of verbs and objects.

For example: "Neha broke the mirror with the hammer". In this example case grammar identify Neha as an agent, mirror as a theme, and hammer as an instrument.

History of NLP

In the year 1960 to 1980, key systems were:

SHRDLU

SHRDLU is a program written by **Terry Winograd** in 1968-70. It helps users to communicate with the computer and moving objects. It can handle instructions such as "pick up the green ball" and also answer the questions like "What is inside the black box." The main importance of SHRDLU is that it shows those syntax, semantics, and reasoning about the world that can be combined to produce a system that understands a natural language.

LUNAR

LUNAR is the classic example of a Natural Language database interface system that is used ATNs and Woods' Procedural Semantics. It was capable of translating elaborate natural language expressions into database queries and handle 78% of requests without errors.

1980 - Current

Till the year 1980, natural language processing systems were based on complex sets of hand-written rules. After 1980, NLP introduced machine learning algorithms for language processing. In the beginning of the year 1990s, NLP started growing faster and achieved good process accuracy, especially in English Grammar. In 1990 also, an electronic text introduced, which provided a good resource for training and examining natural language programs. Other factors may include the availability of computers with fast CPUs and more memory. The major factor behind the advancement of natural language processing was the Internet.

Now, modern NLP consists of various applications, like **speech recognition**, **machine translation**, and **machine text reading**. When we combine all these applications then it allows the artificial intelligence to gain knowledge of the world. Let's consider the example of AMAZON ALEXA, using this robot you can ask the question to Alexa, and it will reply to you.

How Natural Language Processing Works

Simple Example of NLP: “**Ravi is happy with the new phone.**”

An NLP system can:

- Detect Ravi as a person
- Identify phone as an object
- Recognize sentiment as positive
- Understand topic as product review

1. Text or Speech Input

- **Receiving text data:** The system takes written language like sentences or documents which is called text acquisition.
- **Receiving voice input:** When the input is audio, it is first converted into text using Speech Recognition.

2. Pre-processing

The text is cleaned and prepared. It can include:

- **Removing punctuation or noise:** Cleaning unwanted characters or symbols from text is done using text normalization.
- **Splitting into words:** Breaking sentences into smaller units so they can be processed easily.
- **Converting to lowercase:** Changing all words into the same case for uniform processing is known as case folding.
- **Removing common words:** Eliminating frequent words like is, the, and to focus on meaningful terms.
- **Reducing words to base form:** Converting words like running to run to reduce computational power.

How Natural Language Processing Works

3. Language Analysis

The system studies structure and meaning:

- **Grammar detection:** Identifying nouns, verbs, and other parts of speech in a sentence is done.
- **Word relationships:** Finding how words connect to each other in a sentence.
- **Context understanding:** Determining the actual meaning of a word based on surrounding text.
- **Finding names and places:** Detecting entities like person names, locations, or dates.
- **Sentiment detection:** Identifying whether text expresses positive, negative or neutral emotion.

4. Text Representation and Embedding Techniques

Since machines process numbers, this stage converts text into numerical vectors.

- **Text representation:** In this step, text is converted into numbers using statistical features or vector representations so machines can process it.
- **Traditional representations:** Earlier methods represent text using word counts and importance scores.
- **Word embeddings:** Modern methods represent words as dense vectors capturing similarity and meaning.
- **Contextual embeddings:** Advanced models generate word meanings based on the surrounding sentence.

How Natural Language Processing Works

5. Model Training

Once text is numeric, models are trained to learn patterns and perform NLP tasks.

Model training: After text is converted into vectors, algorithms learn patterns from data to perform tasks like classification or translation.

Traditional machine learning: Earlier NLP systems relied on statistical algorithms that learn from manually prepared features.

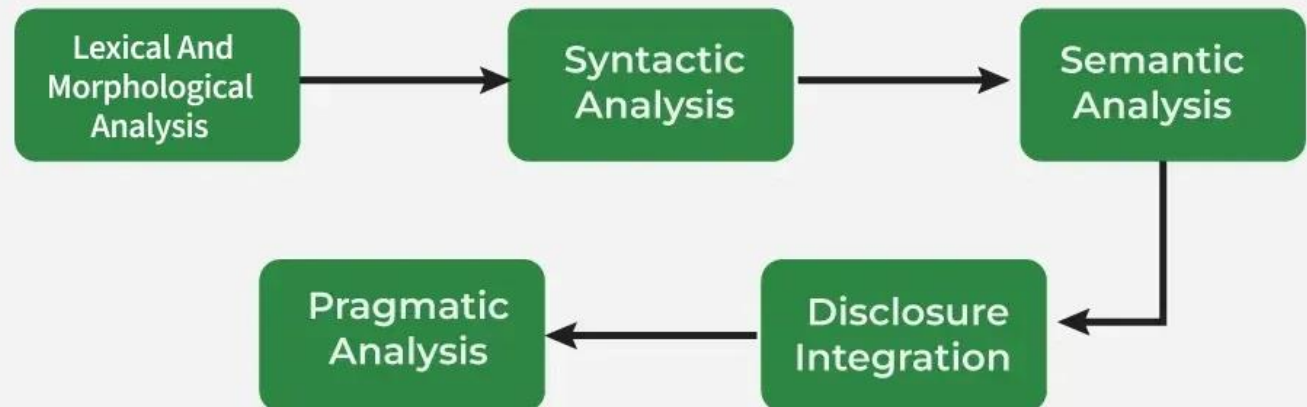
Deep learning approaches: Modern NLP uses neural networks that automatically learn language structure from large data.

Pre-trained models: Large language models trained on massive datasets can be reused and fine-tuned for tasks.

6. Output Generation

The system produces results such as:

- Text reply
- Voice response
- Translation
- Summary
- Prediction



Phases of Natural Language Processing (NLP)

1. Lexical and Morphological Analysis

Lexical and Morphological Analysis are the initial phases of NLP that help computers understand words, their structure and their grammatical roles.

- **Lexical Analysis**

Lexical analysis focuses on identifying and processing words by breaking text into meaningful units called tokens.

Tokenization splits text into smaller units called tokens

Part-of-Speech Tagging assigns grammatical roles such as noun, verb or adjective to words

Helps identify words for further language processing

Simplifies text for NLP tasks

Example

Sentence: "I am reading a book."

Tokenization: ["I", "am", "reading", "a", "book"]

Part-of-Speech Tagging: ["I" → Pronoun (PRP), "am" → Verb (VBP), "reading" → Verb (VBG), "a" → Article (DT), "book" → Noun (NN)]

- **Morphological Analysis**

Morphological analysis studies the internal structure of words using morphemes, the smallest units of meaning.

Stemming reduces words to their root form like "running" to "run".

Lemmatization converts words to their base or dictionary form considering the context like "better" becomes "good". It helps understand word formation and structure

And improves accuracy in parsing, translation and tagging tasks

Supports advanced NLP applications through word normalization

2. Syntactic Analysis (Parsing)

Syntactic Analysis, also called parsing, helps understand how words are arranged in a sentence according to grammar rules. It identifies relationships between words and creates a parse tree representing the sentence structure.

Phases of Natural Language Processing (NLP)

1)Analyzes grammatical structure of sentences, 2)Identifies components such as subject, verb and object 3)Creates parse trees to represent sentence structure 4)Helps machines understand relationships between words

Key components

POS Tagging: Assigns grammatical categories such as noun, verb, and adjective

Ambiguity Resolution: Handles words with multiple meanings based on context

Example

Consider the following sentences

- Correct Syntax: "John eats an apple."
- Incorrect Syntax: "Apple eats John an."

Although both sentences contain the same words, only the first sentence follows correct grammatical structure and conveys meaningful information.

3. Semantic Analysis

Semantic Analysis focuses on understanding the meaning of words and sentences. It ensures that text is not only grammatically correct but also logically meaningful and contextually appropriate.

1)Understands the meaning of words and sentences 2)Checks logical and contextual correctness of text,3)Interprets words based on surrounding context,4)Improves language understanding in NLP systems

Key Tasks

Named Entity Recognition (NER): Identifies entities such as people, locations, organizations and dates

Word Sense Disambiguation (WSD): Determines the correct meaning of words with multiple meanings based on context

Example

Sentence: "Apple eats a John."

Although the sentence is grammatically correct, it is semantically incorrect because an apple cannot perform the action of eating a person.

Phases of Natural Language Processing (NLP)

4. Discourse Integration

Discourse Integration focuses on understanding how multiple sentences connect within a larger context. It ensures that the meaning of a text remains coherent and consistent across sentences and paragraphs.

- 1) Connects related sentences and ideas
- 2) Maintains contextual understanding across text
- 3) Helps interpret references and meanings correctly
- 4) Improves coherence in long or complex texts

Key Aspects

- **Anaphora Resolution:** Identifies references such as pronouns and links them to the correct subject
- **Contextual References:** Understands words or phrases based on surrounding context

Example

"Taylor went to the store. She bought groceries." : "She" refers to "Taylor"

"This is unfair!" : The meaning of "this" depends on the surrounding context

5. Pragmatic Analysis

Pragmatic Analysis focuses on understanding the intended meaning of words and sentences by considering context, tone, and speaker intention beyond the literal meaning.

- Understands implied meanings and intentions

- 1) Interprets context, tone and conversational meaning
- 2) Identifies figurative language such as idioms and metaphors
- 3) Helps machines understand human communication naturally
- 4) Supports sentiment analysis, chatbots, and conversational AI
- 5) Improves context-aware and human like responses

Key Tasks

Understanding Intentions: Identifies the actual purpose behind statements or questions

Figurative Meaning: Interprets non literal expressions such as idioms and metaphors

Examples

"Can you pass the salt?" : A polite request, not a question about ability

"I'm falling for you." : Expresses love, not literal falling

"Hello! What time is it?" : May indicate concern about being late depending on context

Popular NLP (Natural Language Processing) Libraries)

Natural Language Processing (NLP) libraries provide tools for processing, analyzing, and understanding human language data.

Library	Description	Main Features	Example Use Cases
NLTK (Natural Language Toolkit)	One of the oldest and most widely used NLP libraries for education and research.	Tokenization, stemming, lemmatization, POS tagging, parsing, corpora access.	Text preprocessing, sentiment analysis, language learning.
spaCy	Industrial-strength NLP library designed for speed and production use.	Named Entity Recognition (NER), POS tagging, dependency parsing, word vectors.	Chatbots, information extraction, large-scale NLP applications.
TextBlob	Beginner-friendly NLP library built on top of NLTK and Pattern.	Sentiment analysis, noun phrase extraction, translation, POS tagging.	Quick text analysis and prototyping.
Transformers (Hugging Face)	State-of-the-art deep learning library for NLP and generative AI models.	BERT, GPT, T5, RoBERTa, translation, summarization, question answering.	Chatbots, machine translation, text generation, LLM applications.
Gensim	Library specialized for topic modeling and vector-space modeling.	Word2Vec, Doc2Vec, LDA topic modeling.	Document similarity, topic extraction, recommendation systems.

Components of NLP

There are the following two components of NLP -

1. Natural Language Understanding (NLU)

Natural Language Understanding (NLU) is a subfield of Natural Language Processing that provides machines with the ability to interpret and extract meaning from human language by extracting the metadata from content such as concepts, entities, keywords, emotion, relations, and semantic roles. It enables systems to understand the intent behind linguistic inputs. NLU serves as the foundation for a wide range of language-driven applications including chatbots, virtual assistants and content moderation systems.

Key Models and Techniques in NLU

1. **Transformers:** Modern NLU is powered by transformer architectures that capture contextual relationships -

BERT(Bidirectional Encoder Representations from Transformers): BERT is a natural language processing model developed by Google that understands the context of words in a sentence by analyzing text in both directions. Uses bidirectional attention to understand sentence meaning.

T5(Text-to-Text Transfer Transformer): T5 is a transformer-based model developed by Google Research. Treats every task as text-to-text, which simplifies fine-tuning.

GPT(Generative Pre-trained Transformer): GPT is a language model that understands and generates human like text. Focuses on generating and understanding text in a conversational setting.

2. **Recurrent Neural Networks (RNNs):** RNNs analyze text sequentially and maintain context. Variants like **LSTM (Long Short-Term Memory)** and **GRU (Gated Recurrent Unit)** handle long-term dependencies and improve stability.

3. **Word Embeddings:** **Word2Vec** and **GloVe** map words into dense vector spaces where similar meanings lie closer, helping machines reason about semantic similarity.

4. **Rule-Based Systems:** Useful for domain-specific systems, especially in the cases when predictable structure exists.

5. **Conditional Random Fields (CRFs):** CRFs are used in sequence labeling tasks such as POS tagging and NER, capturing dependencies between predicted labels.

Components of NLP

2. Natural Language Generation (NLG)

Natural Language Generation (NLG) acts as a translator that converts the computerized data into natural language representation. It mainly involves Text planning, Sentence planning, and Text Realization. It enables machines to articulate thoughts or observations in natural language.

NLG is the task of generating text that resembles human writing, given a structured or semi-structured input. This input can be in the form of tabular data, knowledge graphs or programmatic events. NLG can be viewed as the reverse of [Natural Language Understanding \(NLU\)](#). While NLU transforms human language into machine-readable formats (parsing questions or commands), NLG transforms machine-represented information into human-friendly text.

How does NLG work

A typical NLG pipeline consists of the following stages:

- **Content Determination:** The system decides which information from the input data is relevant and should be mentioned. This involves filtering facts based on context or importance.
- **Document Structuring:** The content is organized into a continuous structure. Decisions are made about the order in which topics or facts should be presented.
- **Aggregation:** Facts are grouped to improve fluency and reduce redundancy. This ensures the text reads naturally, like how a human would summarize multiple data points.

Components of NLP

- **Lexicalization:** Appropriate words and expressions are chosen to represent the facts.
- **Referring Expression Generation:** The system generates references to entities such as “it”, “they” or proper names to maintain clarity and consistency across the text.
- **Linguistic Realization:** Grammar rules are applied to construct well-formed sentences.

Difference between NLU and NLG

NLU	NLG
NLU is the process of reading and interpreting language.	NLG is the process of writing or generating language.
It produces non-linguistic outputs from natural language inputs.	It produces constructing natural language outputs from non-linguistic inputs.

Merits/Demerits of NLP

Advantages of NLP

NLP helps users to ask questions about any subject and get a direct response within seconds.

- NLP offers exact answers to the question means it does not offer unnecessary and unwanted information.
- NLP helps computers to communicate with humans in their languages.
- It is very time efficient.
- Most of the companies use NLP to improve the efficiency of documentation processes, accuracy of documentation, and identify the information from large databases.

Disadvantages of NLP

A list of disadvantages of NLP is given below:

- NLP may not show context.
- NLP is unpredictable
- NLP may require more keystrokes.
- NLP is unable to adapt to the new domain, and it has a limited function that's why NLP is built for a single and specific task only.

Applications of NLP

There are the following applications of NLP -

1. Question Answering

Question Answering focuses on building systems that automatically answer the questions asked by humans in a natural language.

2. Spam Detection

Spam detection is used to detect unwanted e-mails getting to a user's inbox.

3. Sentiment Analysis

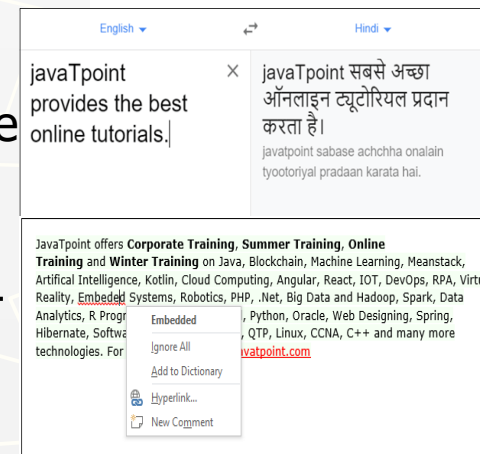
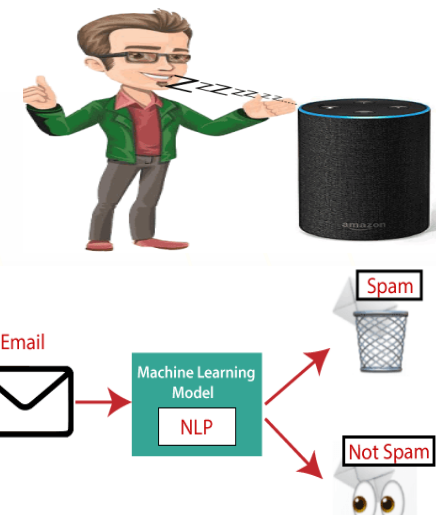
Sentiment Analysis is also known as **opinion mining**. It is used on the web to analyse the attitude, behaviour, and emotional state of the sender. This application is implemented through a combination of NLP (Natural Language Processing) and statistics by assigning the values to the text (positive, negative, or neutral), identify the mood of the context (happy, sad, angry, etc.)

4. Machine Translation

Machine translation is used to translate text or speech from one natural language to another natural language. **Example:** Google Translator

5. Spelling correction

Microsoft Corporation provides word processor software like MS-word, PowerPoint for the spelling correction.



Applications of NLP

6. Speech Recognition

Speech recognition is used for converting spoken words into text. It is used in applications, such as mobile, home automation, video recovery, dictating to Microsoft Word, voice biometrics, voice user interface, and so on.

7. Chatbot

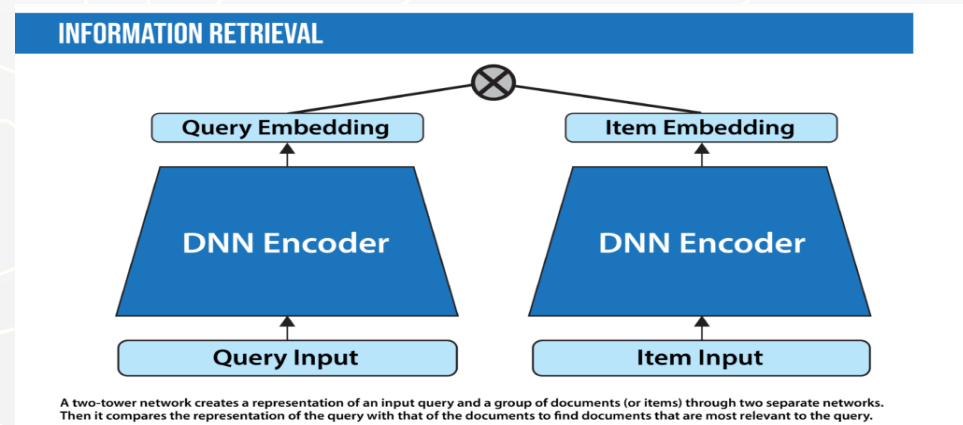
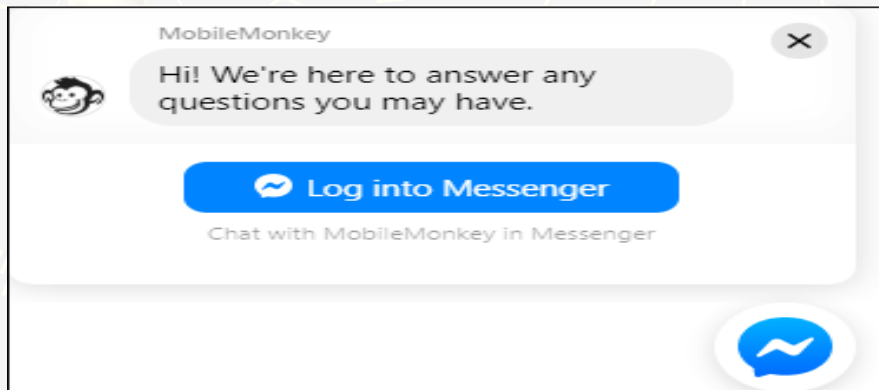
Implementing the Chatbot is one of the important applications of NLP. It is used by many companies to provide the customer's chat services.

8. Information extraction

Information extraction is one of the most important applications of NLP. It is used for extracting structured information from unstructured or semi-structured machine-readable documents.

9. Natural Language Understanding (NLU)

It converts a large set of text into more formal representations such as first-order logic structures that are easier for the computer programs to manipulate notations of the natural language processing.



Applications of NLP

10. Named entity recognition aims to extract entities in a piece of text into predefined categories such as personal names, organizations, locations, and quantities. The input to such a model is generally text, and the output is the various named entities along with their start and end positions. Named entity recognition is useful in applications such as summarizing news articles and combating disinformation. For example, here is what a named entity recognition model could provide:

NAMED ENTITY RECOGNITION (NER) TAGGING

Andrew Yan-Tak Ng **PERSON** (**Chinese** **NORP** : 吳恩達; born **1976** **DATE**) is a **British** **NORP** -born **American** **NORP** computer scientist and technology entrepreneur focusing on machine learning and **AI** **GPE** .
Ng was a co-founder and head of **Google Brain** **ORG** and was the former chief scientist at **Baidu** **ORG** ,
building the company's **Artificial Intelligence Group** **ORG** into a team of **several thousand** **CARDINAL** people.

spaCy named entity recognition tagging of the first paragraph of Andrew Ng's Wikipedia page. "NORP" stands for nationalities or religious or political groups. Note that spaCy incorrectly labels "AI" as "GPE," for geopolitical entity.

11. Toxicity classification is a branch of sentiment analysis where the aim is not just to classify hostile intent but also to classify particular categories such as threats, insults, obscenities, and hatred towards certain identities. The input to such a model is text, and the output is generally the probability of each class of toxicity. Toxicity classification models can be used to moderate and improve online conversations by [silencing offensive comments](#), [detecting hate speech](#), or [scanning documents for defamation](#).

Applications of NLP

12. Summarization is the task of shortening text to highlight the most relevant information. Researchers at Salesforce developed a summarizer that also [evaluates factual consistency](#) to ensure that its output is accurate. Summarization is divided into two method classes:

- **Extractive summarization** focuses on extracting the most important sentences from a long text and combining these to form a summary.
- **Abstractive summarization** produces a summary by paraphrasing. This is similar to writing the abstract that includes words and sentences that are not present in the original text.

13. Question answering deals with answering questions posed by humans in a natural language. One of the most notable examples of question answering was [Watson](#), which in 2011 played the television game-show *Jeopardy* against human champions and won by substantial margins. Generally, question-answering tasks come in two flavors:

Multiple choice:

Open domain

How to build an NLP pipeline

There are the following steps to build an NLP pipeline -

Step1: Sentence Segmentation

Sentence Segment is the first step for building the NLP pipeline. It breaks the paragraph into separate sentences.

Example: Consider the following paragraph -

Independence Day is one of the important festivals for every Indian citizen. It is celebrated on the 15th of August each year ever since India got independence from the British rule. The day celebrates independence in the true sense.

Sentence Segment produces the following result:

"Independence Day is one of the important festivals for every Indian citizen."

"It is celebrated on the 15th of August each year ever since India got independence from the British rule."

"This day celebrates independence in the true sense."

Step2: Word Tokenization

Word Tokenizer is used to break the sentence into separate words or tokens.

Example:JavaTpoint offers Corporate Training, Summer Training, Online Training, and Winter Training.

Word Tokenizer generates the following result:"JavaTpoint", "offers", "Corporate", "Training", "Summer", "Training", "Online", "Training", "and", "Winter", "Training", "."

Step3: Stemming

Stemming is used to normalize words into its base form or root form. For example, celebrates, celebrated and celebrating, all these words are originated with a single root word "celebrate." The big problem with stemming is that sometimes it produces the root word which may not have any meaning.

For Example, intelligence, intelligent, and intelligently, all these words are originated with a single root word "intelligen." In English, the word "intelligen" do not have any meaning.

How to build an NLP pipeline

Step 4: Lemmatization

Lemmatization is quite similar to the Stemming. It is used to group different inflected forms of the word, called Lemma. The main difference between Stemming and lemmatization is that it produces the root word, which has a meaning.

For example: In lemmatization, the words intelligence, intelligent, and intelligently has a root word intelligent, which has a meaning.

Step 5: Identifying Stop Words

In English, there are a lot of words that appear very frequently like "is", "and", "the", and "a". NLP pipelines will flag these words as stop words. **Stop words** might be filtered out before doing any statistical analysis.

Example: He is a good boy.

Step 6: Dependency Parsing

Dependency Parsing is used to find that how all the words in the sentence are related to each other.

Step 7: POS tags

POS stands for parts of speech, which includes Noun, verb, adverb, and Adjective. It indicates that how a word functions with its meaning as well as grammatically within the sentences. A word has one or more parts of speech based on the context in which it is used.

Example: "Google" something on the Internet.

In the above example, Google is used as a verb, although it is a proper noun.

Step 8: Named Entity Recognition (NER)

Named Entity Recognition (NER) is the process of detecting the named entity such as person name, movie name, organization name, or location.

Example: Steve Jobs introduced iPhone at the Macworld Conference in San Francisco, California.

Step 9: Chunking

Chunking is used to collect the individual piece of information and grouping them into bigger pieces of sentences.

Phases of NLP

There are the following five phases of NLP:

1. Lexical Analysis and Morphological

The first phase of NLP is the Lexical Analysis. This phase scans the source code as a stream of character and converts it into meaningful lexemes. It divides the whole text into paragraphs, sentences, and words.

2. Syntactic Analysis (Parsing)

Syntactic Analysis is used to check grammar, word arrangements, and shows the relationship among the words.

Example: Agra goes to the Poonam

In the real world, Agra goes to the Poonam, does not make any sense, so this sentence is rejected by the Syntactic analyzer.

3. Semantic Analysis

Semantic analysis is concerned with the meaning representation. It mainly focuses on the literal meaning of words, phrases, and sentences.

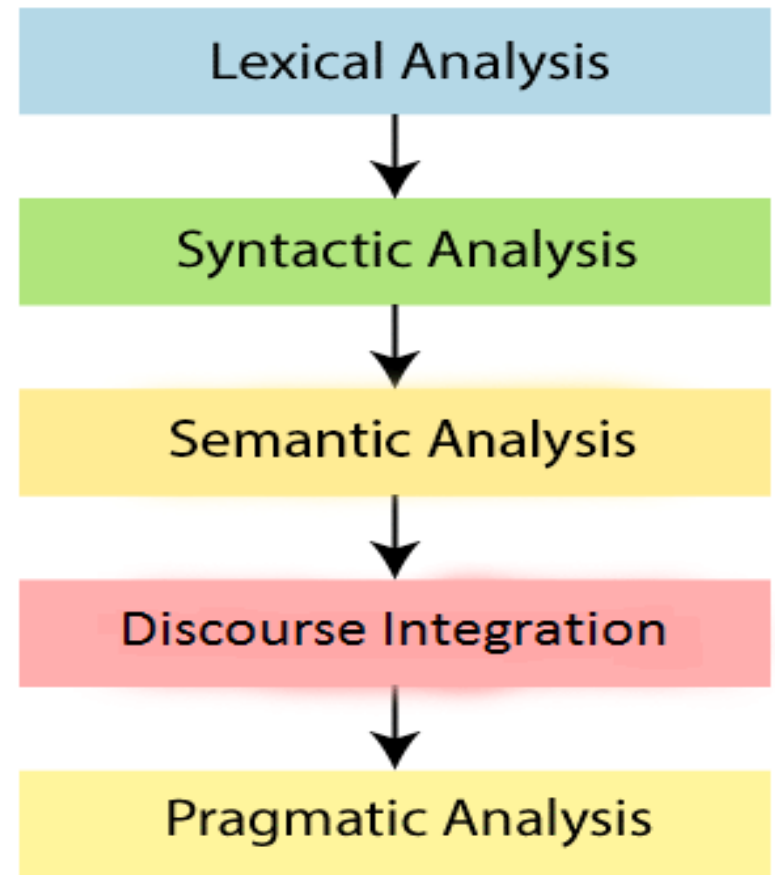
4. Discourse Integration

Discourse Integration depends upon the sentences that proceeds it and also invokes the meaning of the sentences that follow it.

5. Pragmatic Analysis

Pragmatic is the fifth and last phase of NLP. It helps you to discover the intended effect by applying a set of rules that characterize cooperative dialogues.

For Example: "Open the door" is interpreted as a request instead of an order.



Why NLP is difficult?

NLP is difficult because Ambiguity and Uncertainty exist in the language.

Ambiguity

There are the following three ambiguity -

•Lexical Ambiguity

Lexical Ambiguity exists in the presence of two or more possible meanings of the sentence within a single word.

Example:

Manya is looking for a **match**.

In the above example, the word match refers to that either Manya is looking for a partner or Manya is looking for a match. (Cricket or other match)

•Syntactic Ambiguity

Syntactic Ambiguity exists in the presence of two or more possible meanings within the sentence.

Example:

I saw the girl with the binocular.

In the above example, did I have the binoculars? Or did the girl have the binoculars?

•Referential Ambiguity

Referential Ambiguity exists when you are referring to something using the pronoun.

Example: Kiran went to Sunita. She said, "I am hungry."

In the above sentence, you do not know that who is hungry, either Kiran or Sunita.

Important Natural Language Processing (NLP) Models

1. Statistical Models

a) N-Gram Model

- Predicts the next word based on the previous $n-1$ words.
- Types:
 - Unigram
 - Bigram
 - Trigram
- Example:
 - "I am" → predicts "happy"

Applications: Text prediction, speech recognition.

b) Hidden Markov Model (HMM)

- Uses hidden states to model sequential data.
- Commonly used in:
 - Part-of-Speech (POS) Tagging
 - Speech Recognition
 - Named Entity Recognition

c) Naive Bayes Model

- Based on Bayes' theorem.
- Assumes features are independent.

Applications:

- Spam filtering
- Sentiment analysis
- Text classification

Important Natural Language Processing (NLP) Models

2. Word Embedding Models

a) Word2Vec

Developed by Tomas Mikolov.

- Converts words into dense vectors.
- Captures semantic relationships between words.

Techniques:

- CBOW (Continuous Bag of Words)
- Skip-Gram

Example:

- King – Man + Woman $\approx$ Queen

b) GloVe (Global Vectors)

Developed by Stanford University.

- Uses global word co-occurrence statistics.
- Produces meaningful word embeddings.

c) FastText

Developed by Meta.

- Represents words using character-level information.
- Handles rare and misspelled words effectively.

3. Deep Learning Models

a) Recurrent Neural Network (RNN)

- Processes sequential data.
- Maintains information from previous inputs.

Applications:

- Language modeling
- Machine translation

Important Natural Language Processing (NLP) Models

b) Long Short-Term Memory (LSTM)

Improved version of RNN.

Uses memory cells and gates to retain long-term dependencies.

Applications:

Text generation

Sentiment analysis

Speech recognition

c) Gated Recurrent Unit (GRU)

Simplified version of LSTM.

Faster training with comparable performance.

4. Transformer-Based Models

a) Transformer

Introduced in the paper Attention Is All You Need.

- Uses self-attention mechanisms.
- Processes words in parallel.
- Forms the basis of modern NLP systems.

b) BERT (Bidirectional Encoder Representations from Transformers)

Developed by Google.

- Understands context from both directions.
- Excellent for language understanding tasks.

Applications:

- Question answering
- Text classification
- Named Entity Recognition

Important Natural Language Processing (NLP) Models

c) GPT (Generative Pre-trained Transformer)

Developed by [OpenAI](#).

Generates human-like text.

Uses autoregressive language modeling.

Applications:

Chatbots

Content generation

Coding assistants

d) T5 (Text-to-Text Transfer Transformer)

Developed by Google.

Treats every NLP task as a text-to-text problem.

e) RoBERTa

Developed by Meta.

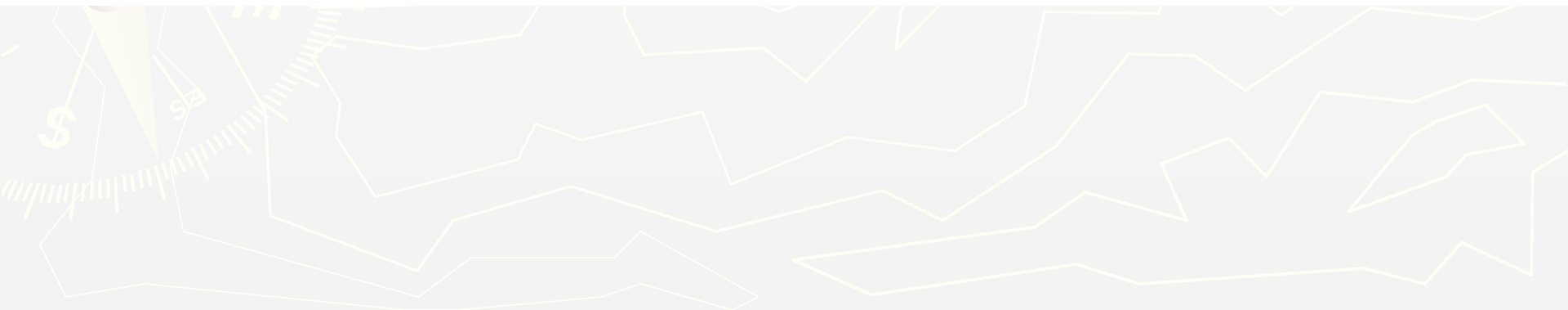
Enhanced version of BERT.

Trained on larger datasets for improved performance.

f) XLNet

Combines advantages of autoregressive and autoencoding approaches.

Achieves strong performance on language understanding tasks.



Top Natural Language Processing (NLP) Techniques

Traditional Machine learning NLP techniques:

Logistic regression

Naive Bayes

Decision trees

Latent Dirichlet Allocation (LDA)

Hidden Markov models

Deep learning NLP Techniques:

Convolutional Neural Network (CNN):

Recurrent Neural Network (RNN)

Autoencoders

Encoder-decoder sequence-to-sequence

Transformers